

Projecting possible future trajectories for systems of cities

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Designing sustainable policies for territorial systems requires an approach encompassing both generic processes and geographical peculiarities of diverse urban systems across the world, in a multi-level way [Rozenblat and Pumain, 2018]. This can be achieved using simulation models for systems of cities [Raimbault and Pumain, 2021]. An evolutionary theory of urban systems has been developed to that end the last 30 years [Pumain, 1997, Pumain and Reuillon, 2017], in parallel to several simulation models for systems of cities [Pumain, 2011]. Methods for exploring various contingencies in such simulation models have been developed in close link to the validation of these models: for example the MARIUS model family coupling population growth with economic exchanges has been challenged into “unexpected” behaviours with the Pattern Space Exploration (PSE) algorithm [Chérel et al., 2015]. Similarly, the SimpopNet model for the co-evolution of transportation infrastructure networks and cities has been generalised into a synthetic setup by [Raimbault, 2020b] to compute its sensitivity to the spatial configuration [Raimbault et al., 2019].

We build in this contribution on such validation methods [Raimbault and Pumain, 2019], to introduce a new composite method for projecting possible future plausible trajectories in simulation models for system of cities. Given a simulation model which can be calibrated on real world data, the method proceeds in three steps: (i) calibrate the model with specified objectives (in practice we follow [Raimbault, 2020a] and use two dimension capturing errors on population trajectories, what produces large sets of solution across Pareto fronts); (ii) from the set of solutions, extract parameter values to obtain plausible parameter ranges for past trajectories; (iii) run the PSE diversity search algorithm on dynamic trajectories a few time steps in the future with the previously obtained parameter bounds.

Our method is applied to a multi-model introduced by [Raimbault and Pumain, 2022] combining several dimensions of urban interactions (infrastructure, economics, innovation) and computing many indicators for Sustainable Development Goals (up to five indicators [Raimbault and Pumain, 2024]). This model is calibrated on population trajectories but can be used to study sustainability issues including proxies for emissions and inequalities, in a very stylised way. We run the model with this model for the European and Chinese system of cities, using harmonised datasets for population trajectories of cities constructed by [Cura et al., 2017], and in particular the ChinaCities database [Swerts, 2017].

First results show that the method is well suited to project future trajectories, with a ensemble of plausible trajectories obtained and the most probable one. In that sense, it provides an alternative to ensemble model approaches or Bayesian approaches. For the systems studied and population trajectories, we obtain less uncertainty in parameters for Europe compared to China, and therefore much broader potential futures for China.

Future work includes (i) the calibration on other dimensions than population, in particular innovation and emissions, to allow a robust interpretability of output future trajectories for these dimensions; (ii) the use of more realistic submodel couplings to include interactions between dimensions (for example innovation reducing emissions); (iii) test of the method with other simulation models and comparison with similar methods.

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